

Flower Turbines Supplementary Instructions for Charge Controllers

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TGET charge controllers are either wind only or hybrid (wind and solar together). The hybrid ones can be used for wind or solar power alone.

The display on the upper middle is where you interface with the controller.

The green section on the side is for wires. From left to right, there are positions for battery, solar, wind, and output. The system requires a battery to keep it operating.



Operations consume around 2.5 watts each hour. Your choice of battery should take into account the amount of charge you want stored during times when there is no energy input from wind or solar.

The controller can work from wind or solar alone, but works best with both.

The controller may have a boost function to increase low voltages to higher ones and/or a buck function to decrease high voltages to lower ones. You do not need to set this.

The following instructions from their manual are good:

2.2.3.1 Battery Wiring

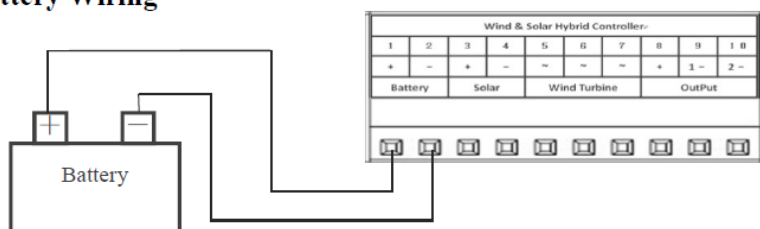


Figure2-2 Battery wiring

Connect battery positive(+) and negative(-) wires to controller as shown in figure2-2.



Be careful of avoiding short circuit when wiring the battery.



Although controller has the protection of battery anti-reverse, but anti-connecting of positive (+) and negative(-) is forbidden.

2.2.3.2 Solar Wiring

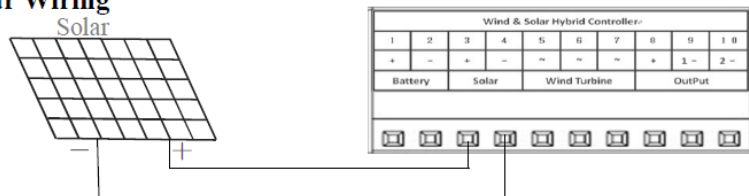


Figure2-3 Solar wiring



Connect solar positive(+) and negative(-) wires to controller as shown in figure2-3.



The solar PV array may produce high voltages in sunlight. Be careful of electric shock when wiring.



Although controller has the protection of solar anti-reverse, but anti-connecting of positive (+) and negative(-) is forbidden.

2.2.3.3 Wind Generator Wiring

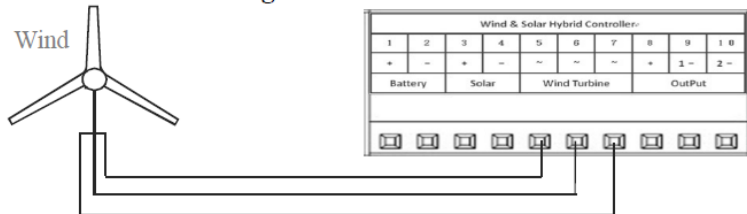


Figure2-4 Wind generator wiring

Connect wind generator wires to controller as shown in figure2-4.



The wind generator could produce high voltages. Be careful of electric shock.

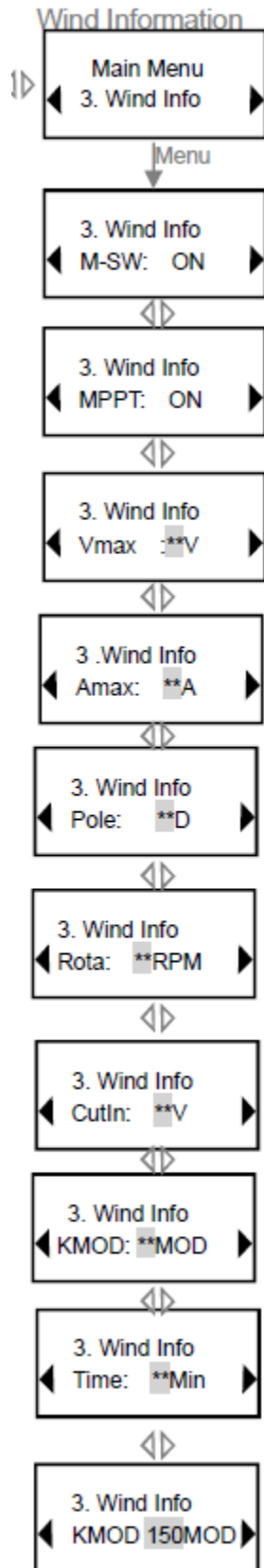


When it is breeze or windless, connection of wind generator and controller would be safer and better. Only when controller is in the state of start-up, high-speed rotate wind generator could be connected.

The button description is good:

Buttons	Description
Menu	Enter into sub-screen or confirm the command.
◀	Switch between sibling menu or decrease the setting value.(Press more than two seconds change the setting value quickly)
▶	Switch between sibling menu or increase the setting value.(Press more than two seconds change the setting value quickly)
Esc	Return to parent screen or cancel the command.

You use the display to navigate to the wind settings:



This is the main wind menu.

Leave M-SW on.

MPPT controls the efficiency of conversion of wind energy to electricity. It uses a power curve that you will enter. In some cases, particularly low wind areas, you may obtain more power by turning MPPT off.

You set the maximal voltage for safety. If the generator is 48 volts, a setting of 48 or slightly above is recommended. (Most generators will operate for some time above their rated voltage capacity, so setting it at 60 will enable you to get power from strong gusts of wind without jeopardizing the generator.

The maximum amps before the turbine shuts off. Usually the factory setting is OK. For a 100 =-watt generator, 10 is fine. For a 500 watt generator, 25 is fine.s

The number of poles depends on the generator. For most 100-watt generators, it is 8. For most 500-watt generators, use 18.

RPM is revolutions per minute. This should match the generator specification. Most 100-watt generators we use are 200 rpm. Most 500-watt generators we use are 100 rpm. This is a braking control.

Cut-in is when the voltage starts being produced. You usually want to set this as low as possible.

KMOD should be set higher than the factory setting. 350 is a good choice.

This is braking time. It's the time the turbine will take to restart after the parameters of the turbine go into a higher speed wind that could be harmful to the electronics. Set Min as low as possible. Usually it is 1-3 minutes.

Some important settings for lower winds especially are setting the Kmod to at least 350 and turning off MPPT and decreasing the minimum voltage.

KMOD refers to the Coefficient of MPPT

The value of KMOD lower: The wind output voltage will be lower. The wind output current will be higher. The value of KMOD higher: The wind output voltage will be higher. The wind output current will be lower. The factory default setting of KMOD is 150. We generally set it at 350, but if you have time, increase it from 150 by increments of 50 and compare its performance.

You can input a power curve to adjust volts and watts for the best efficiency. There are usually 10 settings from low to high, and the others should be spread out in between. Here is an example of settings for a 48-volt, 100-watt generator when it is the only turbine, no cluster effect involved. The voltages and watts picked are not exactly symmetrical because of the logarithmic relationship of wind speed to power.

For different numbers of turbines in a cluster, we will give you guidance as to the recommended parameters.

Setting Number	Volts	Watts
1	0	0
2	0	0
3	1	1
4	8	20
5	17	36
6	25	52
7	32	66
8	38	78
9	44	90
10	48	100

Note that you can vary setting 10 to catch slightly more intense winds.

Here is another example of the settings with voltage increased in the lower part of the curve.

V	6->12	12->24	18->33	24->36	30->40	36->44	42->48	48->52	54->56	60
P in watts	1	3	6	11	18	28	41	56	77	100

In an area of mostly low wind, you don't want to decrease the ability to spin too soon by cutting off the turbine's ability to develop inertia. In that case, one might vary the first few setting

numbers on the power curve as follows so it can reach higher voltages before the harvesting of electricity slows down the spin. The basic principle is that taking power for use reduces the mechanical power available to enable the spinning.

Setting Number	Volts	Watts
1	0	0
2	0	0
3	0	0
4	0	0
5	17	36
6	25	52
7	32	66
8	38	78
9	44	90
10	48	100

A low-wind power curve for a 500-watt, 48-volt generator might be as follows:

Setting Number	Volts	Watts
1	0	0
2	0	0
3	30	0
4	35	10
5	40	40
6	45	80
7	50	130
8	55	200
9	60	300
10	65	500

Setting these curves is complex, and you will likely have to put some experiential time into this to get a perfect result for your particular wind regimen.

When you are programming, The voltage must increase every next setpoint and you have to start with the higher voltages

- “V_{max}” = “70V”
- “Rota” = “220R”

- “Cut In” = “25V”
- “Pole” = “8D”

And put MPPT on and input the power curve according to the attached spreadsheet and the number of generators attached to a particular controller.

Pytes E-Box 48100R